

Literature Review Table — Zero-Day Attack Detection

No.	Title of the Paper	Authors	Journal & Year	Methodology Used	Key Findings	Research Gap
1	IntrusionNet: Deep Learning-Based Hybrid Model for Detection of Known and Zero-Day Attacks	Sarmad Dheyaa Azeez et al.	CMC, 2026	Combines a deep-learning model with an autoencoder to detect known and unseen attacks using UNSW-NB15.	Achieved 98.80% accuracy and 0.985 F1-score .	Does not combine Random Forest, XGBoost, autoencoder and SHAP in one framework.
2	Detecting Zero-Day Web Attacks with an Ensemble of LSTM, GRU, and Stacked Autoencoders	Vahid Babaey, Hamid Reza Faragardi	Computers, 2025	Uses multiple deep-learning and autoencoder models to detect previously unseen web attacks.	Achieved good zero-day detection with a low false-positive rate .	Focused only on web attacks and uses the CSIC 2012 dataset.
3	Federated Active Meta-Learning with Blockchain for Zero-Day Attack Detection in Industrial IoT	Kapil Kumar, Manju Khari	Peer-to-Peer Networking and Applications, 2025	Uses federated learning and blockchain so multiple IoT devices can learn together without directly sharing their data.	Supports zero-day detection while preserving data privacy .	More complex than a centralized intrusion detection system .
4	A Self-Adaptive Intrusion Detection System for Zero-Day Attacks Using Deep Q-Networks	Mouhammd Alkasassbeh et al.	IEEE Access, 2025	Uses a reinforcement-learning model that adapts its detection decisions based on network behaviour.	Achieved 95.9% binary and 92.0% multiclass detection accuracy.	Evaluated on only one dataset and focuses on ransomware attacks .
5	Explainable AI for Zero-Day Attack Detection in IoT Networks Using Attention Fusion Model	Deepa Krishnan, Swapnil Singh, Vijayan Sugumaran	Discover Internet of Things, 2025	Combines deep learning with explainability techniques to detect and explain zero-day attacks.	Provides zero-day detection with interpretable predictions .	Focused on IoT networks and uses a different explainability approach.
6	Systematic Review of Current Approaches and Innovative Solutions for Combating Zero-Day Vulnerabilities and Zero-Day Attacks	Kübra Nilgün Karaca, Aydın Çetin	IEEE Access, 2025	Reviews 59 research studies on zero-day detection methods.	Identifies the major approaches and research trends in zero-day detection.	It is a review study and does not propose or test a new detection framework.
7	F-OSFA: A Fog Level Generalizable Solution for Zero-Day DDoS Attacks Detection	Muhammad Rashid Minhas et al.	IEEE Access, 2025	Combines deep learning, decision trees and an autoencoder to detect previously unseen DDoS attacks.	Achieved 96.77% / 95.98% accuracy for zero-day DDoS detection.	Focused specifically on DDoS attacks and fog computing .
8	ZeroCAN: Anomaly-Based Zero-Day Attack Detection in Vehicular CAN Bus Networks	Jonathan Rendel et al.	IEEE Euromicro PDP, 2025	Uses anomaly detection to identify unusual behaviour in vehicle communication networks.	Achieved over 99% anomaly detection with a low false-positive rate.	Limited to vehicular CAN networks , not general network intrusion detection.